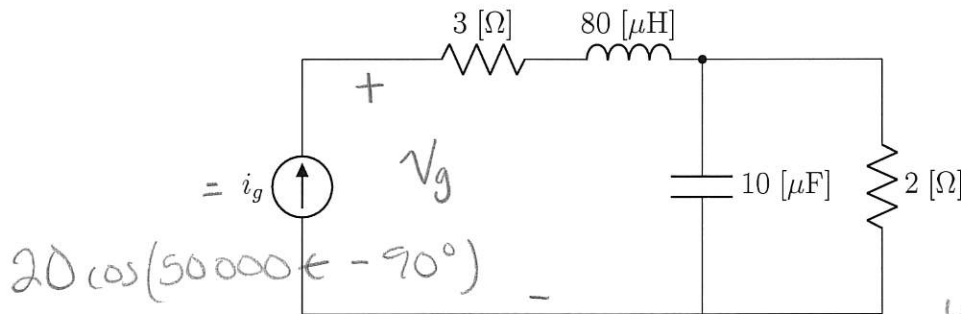


30

{30 Points} The circuit shown operates in steady-state. The ideal current source has $i_g(t) = 20 \sin(50,000 t)$ [mA].



$$\omega = 50000 \text{ [rad/s]}$$

- (5 pts) Find the rms value I_{rms} of the current source.
- (5 pts) Find the impedance of the capacitor Z_C and the impedance of the inductor Z_L .
- (5 pts) Find the equivalent impedance Z_{eq} seen from the current source.
- (5 pts) Find the voltage across the current source (rms phasor).
- (10 pts) Find the complex power S delivered by the ideal current source.

$$I_{g,\text{rms}} = \frac{20 [\text{mA}] \angle -90^\circ}{\sqrt{2}} = 14.14 [\text{mA}_{\text{rms}}] \angle -90^\circ \quad \text{(a)}$$

$$Z_C = -j \frac{1}{\omega \cdot 10 \cdot 10^{-6} [\text{F}]} = -j2 [\Omega] \quad \text{(b)}$$

$$Z_L = j\omega \cdot 80 \cdot 10^{-6} [\text{H}] = j4 [\Omega]$$

$$Z_{\text{eq}} = 3 [\Omega] + j4 [\Omega] + (-j2 [\Omega] \parallel 2 [\Omega]) = 4 + j3 [\Omega] \quad \text{(c)}$$

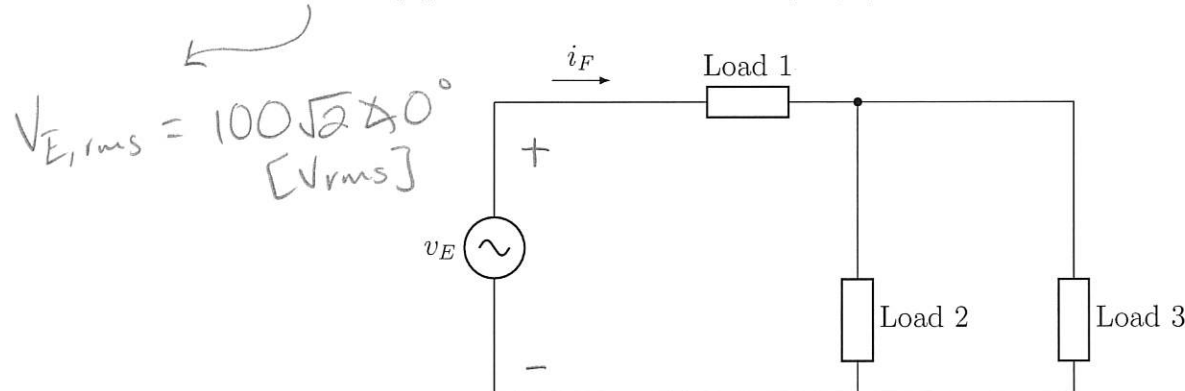
$$V_{g,\text{rms}} = I_{g,\text{rms}} Z_{\text{eq}} = 70.71 [\text{mV}_{\text{rms}}] \angle -53.13^\circ \quad \text{(d)}$$

Room for extra work

$$S_{\text{DEL } i_g} = V_{g,\text{rms}} I_{g,\text{rms}}^* = 1 [\text{mVA}] \angle 36.87^\circ \quad \text{(e)}$$

Extra space for Problem 1.

{30 Points} The circuit shown operates in steady-state. The ideal voltage source has $V_E = 200\angle 0^\circ$ [V] peak value with $\omega = 5000$ [rad/s].



Load 1 absorbs 40 [W] and delivers 30 [VAR].

Load 2 absorbs $(80\angle -25^\circ)$ [VA].

Load 3 absorbs 60 [VA] at a lagging power factor of 0.80.

Lagging: $\theta_{L3} > 0$, $\theta_{L3} = \cos^{-1}(0.80) = 36.87^\circ$

a) (10 pts) Find the complex power delivered by the v_E voltage source.

b) (10 pts) Find the numerical expression for the current $i_F(t)$.

c) (10 pts) Find the impedance of Load 1.

$$S_{ABS L1} = 40 - j30 \text{ [VA]}$$

$$S_{ABS L2} = 80\angle -25^\circ \text{ [VA]}$$

$$S_{ABS L3} = 60\angle 36.87^\circ \text{ [VA]}$$

$$S_{DEL VE} = S_{ABS L1} + S_{ABS L2} + S_{ABS L3} = 162.9\angle -9.83^\circ \text{ [VA]} \quad (a)$$

$$S_{DEL VE} = V_{E,rms} \cdot I_{F,rms}^* \rightarrow I_{F,rms} = \left(\frac{S_{DEL VE}}{V_{E,rms}} \right)^* = 1.152\angle 9.83^\circ \text{ [A}_{rms}]$$

$$I_F = \sqrt{2} I_{F,rms} = 1.629\angle 9.83^\circ \text{ [A]} \rightarrow$$

$$i_F(t) = 1.629 \text{ [A]} \cos(5000t + 9.83^\circ) \quad (b)$$

$$S_{ABS L1} = |I_{F,rms}|^2 Z_{L1} \rightarrow Z_{L1} = \frac{S_{ABS L1}}{|I_{F,rms}|^2} = 30.14 - j22.61 \text{ [}\Omega\text{]} \quad (c)$$

Room for extra work

Extra space for Problem 2.

{40 Points}

A balanced three-phase Y-connected generator with a positive phase sequence (abc) has an internal impedance of $(0.5 + j2) [\Omega/\phi]$. The magnitude of the phase voltage of the generator is $12,000 [V_{rms}]$. The generator feeds two loads in parallel through a transmission line with an impedance of $(1.5 + j6) [\Omega/\phi]$. Load 1 is Δ -connected with a per-phase impedance of $(120 - j90) [\Omega]$. Load 2 is Y-connected with a per-phase impedance of $(60 + j45) [\Omega]$.

- a) (10 pts) Draw the single-phase (a-phase) equivalent circuit. Show all impedance values on your diagram.
- b) (10 pts) Calculate the Y-equivalent impedance of Load 1 and the equivalent per-phase load impedance (Y-equivalent of Load 1 in parallel with Load 2).
- c) (5 pts) Calculate the magnitude of the line current $|I_{aA}|$.
- d) (5 pts) Calculate the magnitude of the line voltage at the terminals of the loads, $|V_{AB}|$.
- e) (5 pts) Calculate the total complex power delivered by the three-phase source.
- f) (5 pts) Calculate the total power lost in the transmission line.

Positive abc: $\phi_{abc} = \{0^\circ, -120^\circ, +120^\circ\}$

$$Z_{g,Y} = 0.5 + j2 [\Omega/\phi]$$

$$|V_{a'n}| = 12,000 [V_{rms}]$$

$$Z_{line} = 1.5 + j6 [\Omega/\phi]$$

$$Z_{L1,\Delta} = 120 - j90 [\Omega/\phi]$$

$$Z_{L2,Y} = 60 + j45 [\Omega/\phi]$$

$$Z_{L1,Y} = \frac{Z_{L1,\Delta}}{3} = 40 - j30 [\Omega/\phi]$$

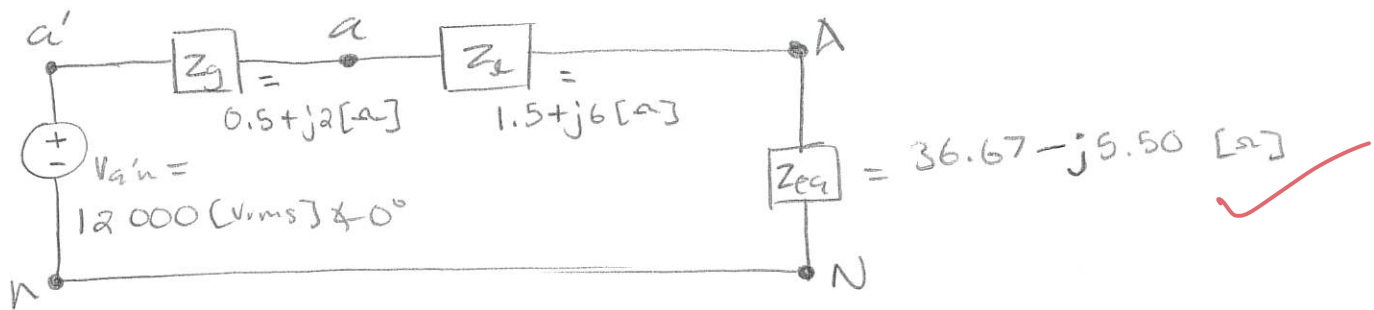
$$Z_{eq,Y} = Z_{L1,Y} \parallel Z_{L2,Y} = 36.67 - j5.50 [\Omega/\phi]$$

Room for extra work

All values in rms!

Single-phase equivalent

Extra space for Problem 3.



$$I_{aA} = V_{a'n} / (Z_g + Z_L + Z_{eq}) = 309.7 [A_{rms}] \angle -3.7^\circ$$

$$|I_{aA}| = 309.7 [A_{rms}] \quad \checkmark \quad (c)$$

$$V_{AB} = V_{AN} - V_{BN} = V_{AN} [1 - (1 \angle -120^\circ)] = V_{AN} (\sqrt{3} \angle 30^\circ)$$

$$V_{AB} = I_{aA} Z_{eq} (\sqrt{3} \angle 30^\circ) = 19889 [V_{rms}] \angle 17.8^\circ$$

$$|V_{AB}| = 19889 [V_{rms}] \quad \checkmark \quad (d)$$

$$S_{DEL \text{ SOURCE } / \phi} = V_{a'n} \cdot I_{aA}^* = 3.716 \angle 3.7^\circ [MVA]$$

$$S_{DEL \text{ SOURCE}} = 3 \cdot S_{DEL \text{ SOURCE } / \phi} = 11.15 \angle 3.7^\circ [MVA] \quad \checkmark \quad (e)$$

$$S_{ABS \text{ LINE } / \phi} = |I_{aA}|^2 Z_L = 593.2 \angle 75.96^\circ [kVA]$$

$$P_{ABS \text{ LINE}} = 3 \cdot \text{Re}(S_{ABS \text{ LINE } / \phi}) = 431.6 [kW]$$

$$\text{Total power lost in lines} = 431.6 [kW] \quad \checkmark \quad (f)$$